

COMMON SHOULDER (& UPPER LIMB) PATHOLOGY

Shoulder impingement syndrome

- **Mechanical compression of rotator cuff (mainly supraspinatus)** + subacromial bursa + long head biceps. Pain on overhead movements.
- **Clinical: painful arc (60–120°)**, night pain, weak abduction.
- **Special tests: Neer** (pain on forced flexion), **Hawkins-Kennedy** (pain on internal rotation), **empty can** (supraspinatus weakness).
- **Management:** conservative (activity modification, NSAIDs, physio = cuff strengthening, subacromial steroid injection) → surgical (**arthroscopic subacromial decompression**, remove spur, repair cuff tear).

Frozen shoulder (adhesive capsulitis)

- **Self-limiting painful stiffness** — chronic inflammation + fibrosis of joint capsule. **Loss of active AND passive ROM (mainly external rotation).**
- Age 40–50; **females**; **strong DM association (2–30%)**. Idiopathic or secondary (trauma, surgery, immobilization).
- **Pathophysiology:** synovial inflammation → fibroblast proliferation → **excess type III collagen** → **capsular thickening** (rotator interval, coracohumeral ligament, axillary pouch).
- **Clinical:** gradual painful stiffness, night pain; **loss of passive external rotation = hallmark.**
- **3 phases: Freezing (painful, 2–9mo) → Frozen (stiff, 4–12mo) → Thawing (6–24mo).**
- **Management:** clinical diagnosis (X-ray rules out arthritis); NSAIDs, physio (stretching), **intra-articular steroid (early)**, MUA / arthroscopic release if refractory.

Calcific tendinitis

- **Calcium hydroxyapatite deposits in rotator cuff (mainly supraspinatus).**
- **Etiology:** tendon hypoxia → fibrocartilaginous metaplasia → calcium deposition. Assoc DM + thyroid disease.
- **4 phases:** formative → resting → **resorptive (severe pain)** → post-calcific.
- **Clinical:** sudden severe pain, night pain, painful arc. **Imaging:** X-ray (dense circumscribed calcification), US, MRI.
- **Management:** NSAIDs, physio, subacromial steroid, **US-guided barbotage, shock-wave therapy**; surgical (arthroscopic removal).

Tennis elbow (lateral epicondylitis)

- **Overuse of ECRB tendon at lateral epicondyle** (repetitive wrist extension + gripping).
- **Clinical:** lateral elbow pain, worse with gripping/lifting, tender over lateral epicondyle, pain on resisted wrist extension.
- **Special tests: Cozen's** (resisted wrist extension → pain), **Mill's** (passive wrist flexion → pain), **Maudsley** (resisted middle finger extension → pain).
- **Management:** rest, NSAIDs, physio (eccentric strengthening), counterforce brace, ESWT, PRP injection; **surgery only if >6–12 months conservative failure.**

Congenital upper limb deformities

- **Radial longitudinal deficiency:** failed radial development → radial deviation. Assoc **VACTERL, Holt-Oram, Fanconi anaemia**. Bayne I–IV. Tx: splinting, centralization/radialization, **pollicization**.

- **Syndactyly**: failed digital separation. Assoc **Apert, Poland**. Surgical release 6–12 months.
- **Polydactyly**: extra digits. Assoc Ellis-van Creveld, Patau. Preaxial/postaxial/central. Excision (Wassel for thumb).
- **Thumb hypoplasia**: underdeveloped thumb. Assoc Holt-Oram, VACTERL. Blauth I–V. Opponensplasty/pollicization.
- **Macroductyly**: digit enlargement. Assoc Proteus, NF1. Debulking, epiphysiodesis.

High-yield one-liners

- **Impingement**: painful arc 60–120°; Neer + Hawkins-Kennedy + empty can tests; subacromial decompression.
- **Frozen shoulder**: loss of **PASSIVE** external rotation = hallmark; DM association; freezing/frozen/thawing phases.
- **Calcific tendinitis**: hydroxyapatite in supraspinatus; resorptive phase = severe pain; barbotage.
- **Tennis elbow** = ECRB; Cozen's/Mill's/Maudsley; eccentric strengthening.
- **Radial deficiency** = VACTERL/Fanconi → pollicization; syndactyly = Apert/Poland.

COMMON SHOULDER & UPPER LIMB PATHOLOGY — Revision Table

Bold = high-yield. Dr Saywan Asaad.

SHOULDER & ELBOW SOFT TISSUE		
Condition	Clinical / tests	Management
Impingement	Painful arc 60–120°, night pain; Neer, Hawkins-Kennedy, empty can	NSAIDs, physio, subacromial steroid → arthroscopic decompression
Frozen shoulder	Loss of passive ER = hallmark; DM assoc ; freezing/frozen/thawing	NSAIDs, physio, intra-articular steroid (early) , MUA if refractory
Calcific tendinitis	Hydroxyapatite in supraspinatus; resorptive phase = severe pain ; X-ray dense calcification	NSAIDs, barbotage, shock-wave ; arthroscopic removal
Tennis elbow	ECRB ; lateral pain, resisted wrist extension; Cozen's/Mill's/Maudsley	Rest, eccentric strengthening, brace, PRP; surgery if >6–12mo

CONGENITAL UPPER LIMB	
Deformity	Key points
Radial longitudinal deficiency	VACTERL, Holt-Oram, Fanconi ; Bayne I–IV; pollicization
Syndactyly	Failed separation; Apert, Poland ; release 6–12mo
Polydactyly	Extra digits; Ellis-van Creveld, Patau; excision (Wassel for thumb)
Thumb hypoplasia	Holt-Oram, VACTERL; Blauth I–V; opponensplasty/pollicization
Macroductyly	Proteus, NF1 ; debulking, epiphysiodesis